Event Generation with Parallel Langevin Sampling and Learned Stein Diagnostics

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Abstract

Efficient event generation is a major computational challenge for precision collider simulations, especially for high-multiplicity final states where matrix-element evaluations are expensive and rejection-sampling efficiencies are low. We study an alternative approach based on many parallel underdamped Langevin chains, retaining one terminal state from each chain to obtain unweighted events while avoiding within-chain autocorrelation. A learned Stein discrepancy is used as a convergence diagnostic, providing a data-driven estimate of the relaxation time. We apply the method to tree-level $u\bar{u} \to Z + ng$ event generation and find that relaxation requires only a modest number of exact-target Langevin steps, with mild growth over the multiplicities studied. Finally, we show that simple neural-network surrogate initialization can substantially reduce the required number of exact matrix-element and gradient evaluations.

1 Introduction

Monte Carlo event simulation is central to particle physics, connecting theoretical predictions with experimental measurements. The high-luminosity upgrade to the Large Hadron Collider (LHC) [1] and its potential successors [2, 3] will demand unprecedented precision and correspondingly large simulated samples, whose production is expected to become a dominant computational bottleneck [4–8].

The most expensive component of the simulator chain [9] is the sampling of the hard proton-proton collision differential cross section

$$\mathrm{d}\sigma_{pp\to X_n} = \frac{1}{\mathcal{F}} \sum_{i,j} f_i(x_1)\, f_j(x_2)\, |M_{ij\to X_n}(\Phi_n)|^2 \,\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}\Phi_n, \tag{1}$$

where the sum runs over the constituents of the proton (quarks, antiquarks and gluons), f are their corresponding parton density functions, $|M_{ij\to X_n}|^2$ is the squared matrix element, and $\mathcal{F}$ is the incoming partonic flux factor. The integration measure is given by the two partonic momentum fractions $\mathrm{d}x_{1,2} \in [0, 1]$ and the n-body phase-space measure $\mathrm{d}\Phi_n$, which integrates over the final-state momenta while ensuring momentum conservation. The total sampling dimensionality of this object is $3n - 2$.

For large multiplicities n and for high theoretical precision, the evaluation of the matrix element can be prohibitively expensive. The method used to sample Eq. (1) must therefore be as efficient as possible in terms of evaluations, even if this comes at increased algorithmic cost. Current methods are still largely based on the VEGAS algorithm [10, 11] with many physics-inspired improvements [12–19]. These methods all rely on an adaptive approximation of an unnormalized target density $\widetilde{p}(x)$ (the differential cross section) by some density $q(x)$ that is easily sampled. One can then sample from

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$q(x)$ and assign weights $w(x) = \widetilde{p}(x)/q(x)$. To obtain unweighted samples, rejection sampling must be performed with acceptance probability

$$p_{\texttt{acc}}(x) = \frac{w(x)}{\max_x w(x)}. \tag{2}$$

The unweighting efficiency $\epsilon = \mathbb{E}_{x\sim q}\left[p_{\texttt{acc}}(x)\right]$ then determines the required number of matrix element evaluations per generated sample. Eq. 2 is very sensitive to the value of $\max_x w(x)$, which may originate from tiny regions of phase space. Thus, to improve efficiency, one may make use of a reduced maximum and leave some undesirable weighted events [20]. Nevertheless, for increasing multiplicities, the unweighting efficiency can still degrade by several orders of magnitude [21].

Recent work has used modern machine-learning (ML) architectures such as normalizing flows in place of VEGAS to improve efficiencies [22–29]. These approaches are however still ultimately based on rejection sampling and thus inherit its disadvantages. In this work, we propose a different approach based on parallelized Markov chain Monte Carlo (MCMC) supplemented by an ML-based convergence diagnostic. MCMC methods have previously been explored for hard-process phase-space sampling and event generation [30–33]. Our method places particular emphasis on avoiding within-chain autocorrelations and on reaching MCMC stationarity, i.e. samples being representative of $\widetilde{p}(x)$. Furthermore, we investigate how neural network surrogates of the target distribution can further reduce the required number of matrix-element evaluations [34–36].[1]

2 Method

We now describe our sampling construction, targeting an unnormalized density $\widetilde{p}(x)$ on the unit hypercube $x \in [0,1]^d$. To avoid within-chain autocorrelation, we evolve many Markov chains in parallel on hardware accelerators and retain only the terminal state from each chain. The learned Stein discrepancy introduced below is used to determine the relaxation time t_{rel}, i.e. the number of MCMC steps after which the ensemble is representative of the target.

2.1 Underdamped Langevin Algorithm

Our MCMC framework is designed for rapid convergence under two practical constraints: the sampler should be straightforward to run over many parallel chains, and it should use expensive target-density and gradient evaluations sparingly. Sophisticated Hamiltonian Monte Carlo (HMC) methods [37] like NUTS [38] implement adaptive trajectory lengths which leads to complex control flow and thus difficult parallelizability. Moreover, HMC typically requires several intermediate gradient evaluations before a single Metropolis-Hastings (MH) correction is applied. We therefore use an Underdamped Langevin Algorithm (ULA) [39, 40], which retains momentum between steps while applying an MH correction after each proposal. Each parallel chain evolves a position–momentum pair using a leapfrog proposal, a MH correction, partial momentum refreshment, and adaptive step sizes. The complete update and adaptation scheme is given in Appendix A.

2.2 Mirror Langevin Dynamics

After the application of a mapping, the phase-space coordinates used for the cross-section integrand live on the bounded domain $x \in [0,1]^d$, while ULA is most naturally formulated on $\mathbb{R}^d$. We therefore perform the MCMC evolution in mirror space [41], in which Langevin dynamics evaluated through a suitable mirror map retain the convergence properties of the corresponding unconstrained algorithm. We use the element-wise transform $y_i = \Phi^{-1}(x_i) = \sqrt{2}\,\mathrm{erf}^{-1}(2x_i - 1)$. The mirrored target density is the corresponding change of variables

$$\log \widetilde{p}_{\mathrm{mir}}(y) = \log \widetilde{p}(\Phi(y)) - \frac{1}{2}\|y\|^2 + \mathrm{const.} \tag{3}$$

Note that the constant drops out in the score.

[1] Code, trained models, and numerical results are available at https://github.com/rbvh/ula_stein_evt_gen.

2.3 Learned Stein Discrepancy

To monitor convergence of the ULA chains, we need a distributional discrepancy that can be evaluated using samples from the chain and the unnormalized density of the target. The Stein discrepancy provides such a diagnostic [42, 43]. For a vector-valued critic function $f : \mathbb{R}^d \to \mathbb{R}^d$, the Stein discrepancy between a smooth, sampled distribution p and a smooth target distribution q is

$$S(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x\sim p} \left[s_q(x)^T f(x) + \nabla \cdot f(x) \right], \tag{4}$$

where $\mathcal{F}$ is a norm-bounded class of differentiable functions for which the expectation over p is finite and $q(x)f(x)$ has vanishing tails. For sufficiently rich critic classes, this quantity is non-negative and satisfies $S(p, q) = 0$ if and only if $p = q$ [44].

Kernel Stein discrepancies are a popular choice for $\mathcal{F}$ [45–47], but their standard estimators involve pairwise sums that scale as n^2 in the number of samples n. Since we typically work in a regime with large numbers of samples, n^2 evaluation is infeasible, and a more natural choice would be one that exploits this large data size. We thus use the learned Stein discrepancy (LSD) [48, 49], taking $\mathcal{F}$ to be a class of square-integrable differentiable functions that satisfy the Stein boundary condition, and representing the critic by a neural network f_ϕ. In practice, square integrability is enforced by adding a quadratic regularization term to the training objective,

$$\mathcal{L}_\lambda(\phi; p, q) = -\mathbb{E}_{x\sim p} \left[s_q(x)^T f_\phi(x) + \nabla \cdot f_\phi(x) - \lambda \| f_\phi(x) \|^2 \right]. \tag{5}$$

After training, the learned Stein discrepancy is evaluated with the learned critic on a held-out sample, without the regularization term.

2.4 Sharp Boundaries

Physical cross sections typically involve sharp phase-space cuts, which restrict the target density to an allowed region with a hard boundary. The derivation of Eq. 4 involves an integration by parts over this region, and a boundary contribution will thus remain unless the Stein test function has vanishing flux through the cut surface [50, 51]. We address this issue by introducing a scalar boundary function $h(x)$ that is positive inside the allowed region and vanishes on the cut boundary. The Stein operator is then applied to the vector field hf rather than to f itself [50, 52],

$$S_h(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x\sim p} \left[h(x) s_q(x)^T f(x) + h(x) \nabla \cdot f(x) + \nabla h(x)^T f(x) \right]. \tag{6}$$

2.5 Neural Surrogates

The sampling strategy described above relies on gradients of the target density. Differentiable matrix element implementations have recently become available [29, 53–56], enabling the sampling strategy set out above, but at higher computational expense. This motivates us to consider the use of learned surrogates to reduce the required number of evaluations.

Neural surrogates for scattering amplitudes and matrix elements have been studied extensively [34–36, 57–71]. In the present setting, one can train a surrogate on a relatively limited set of points without requiring gradients. Then, chains can be evolved cheaply through the early part of the trajectory using the surrogate as target, before switching over to a shorter sequence of ULA steps using the exact target. The surrogate is thus only used to accelerate equilibration, while the final evolution remains tied to the physical target density. We investigate this strategy in Section 3.

3 Results

We validate the learned Stein diagnostic on analytically tractable Ornstein–Uhlenbeck targets in Appendix B.

We apply ULA and the LSD diagnostic to sample the cross section for the process $u\bar{u} \to Z + ng$. Matrix elements and their gradients are evaluated using MadJax [53] with several modifications for numerical stability. We use the NNPDF4.0 LO set [72], and implement the PDF and α_s interpolation in JAX [73] following the LHAPDF interpolation prescription [74]. We set $\sqrt{s} = 13\,\mathrm{TeV}$ and apply cuts $p_\mathrm{T} > 20\,\mathrm{GeV}$ ($n \geq 1$) and $|y| < 5$ to all final-state particles and a separation cut $\Delta R > 0.4$ to

the gluons. Phase space is mapped to the $[0,1]^{3n+1}$ hypercube using the Chili [75] vector boson plus jets map. The Stein diagnostic uses the same critic architecture and optimization strategy as in the OU validation, with $\lambda = 0.1$. The boundary function is

$$h(x) = \left[\frac{1}{2 + \binom{n}{2}} \left(\frac{1}{1 - x_a} + \frac{1}{1 - x_b} + \sum_{1 \le i < j \le n} \frac{1}{1 - \Delta R_{\mathrm{cut}}^2 / \Delta R_{ij}^2} \right) \right]^{-1}, \tag{7}$$

for points that pass the cuts, and 0 for points that do not.

We use ULA and the LSD diagnostic to generate 10^7 samples for $n = 0, 1, 2, 3$, both with and without neural surrogates. The surrogate is a residual MLP with five hidden layers of width 128, GeLU activations, and a scalar output. It is trained to predict the log cross section from only 10^6 samples using Adam and a mean-squared error loss. We then run ULA on the surrogate for a large number of steps. The terminal states are then used as the initial states for the ULA run, which is otherwise initialized with unit Gaussian samples. We define the relaxation time t_{rel} as the first checkpoint for which the learned LSD is no more than one standard error above zero, $S_{\mathrm{LSD}} \le \Delta S_{\mathrm{LSD}}$. After the trigger, we evolve the chain for a further $\max(0.5\, t_{\mathrm{rel}}, 50)$ ULA steps. Figure 1 shows the convergence results, and relaxation-time triggers are summarized alongside VEGAS unweighting results in Table 1. The surrogate warm start substantially reduces the number of exact-target ULA steps required for relaxation. All experiments in this section were run on a single NVIDIA GeForce RTX 5090 GPU (32 GB) and required less than 24 hours in total.

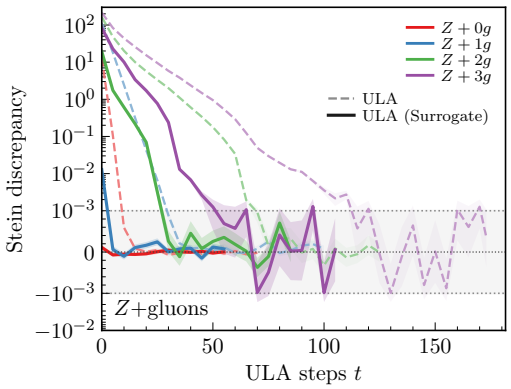


Figure 1: The LSD as a function of the number of ULA steps for $Z + ng$ cross sections. Colours denote gluon multiplicities, while line styles distinguish ULA with and without surrogate initialization.

We compare with VEGAS, reporting $1/\epsilon$, i.e. the number of weighted events per unweighted event. Following standard practice, we also report $1/\epsilon_{0.001}$ and $1/\epsilon_{0.01}$, in which overweights contributing the quoted fractions to the total integral are allowed to remain.

Table 1: Exact-target ULA relaxation-time checkpoints and VEGAS inverse unweighting efficiencies for the same cross sections. The VEGAS uncertainties are standard errors over 10 independent surveys of each fitted grid.

	ULA t_{rel}		VEGAS		
Process	Standard	Surrogate	$1/\epsilon$	$1/\epsilon_{0.001}$	$1/\epsilon_{0.01}$
$Z + 0g$	20	5	$2.69 \pm 5.5 \times 10^{-5}$	$2.53 \pm 8.2 \times 10^{-4}$	$1.25 \pm 2.5 \times 10^{-4}$
$Z + 1g$	50	5	20.0 ± 0.49	8.75 ± 0.029	$4.31 \pm 2.2 \times 10^{-3}$
$Z + 2g$	75	35	540 ± 140	74.8 ± 2.3	30.0 ± 0.085
$Z + 3g$	115	55	1450 ± 240	460 ± 23	148 ± 1.3

We observe that the VEGAS proposal requirement increases substantially with gluon multiplicity, while Langevin relaxation time grows much more mildly. Allowing residual weights improves efficiencies considerably, but leaves some events with non-unit weights, whereas the MCMC construction directly produces unit-weight events. We note that methods like VEGAS benefit greatly from techniques like multichanneling [12], and a more comprehensive comparison should be performed in the context of a full event generator.

Figure 2 compares ULA samples with and without surrogate initialization against independent VEGAS estimates for four physical observables. The jet-resolution scale $\sqrt{d_{01}}$ is obtained using exclusive k_t clustering with $R = 1$ [76, 77]. In every case, the terminal ULA samples reproduce the VEGAS distributions within statistical uncertainties.

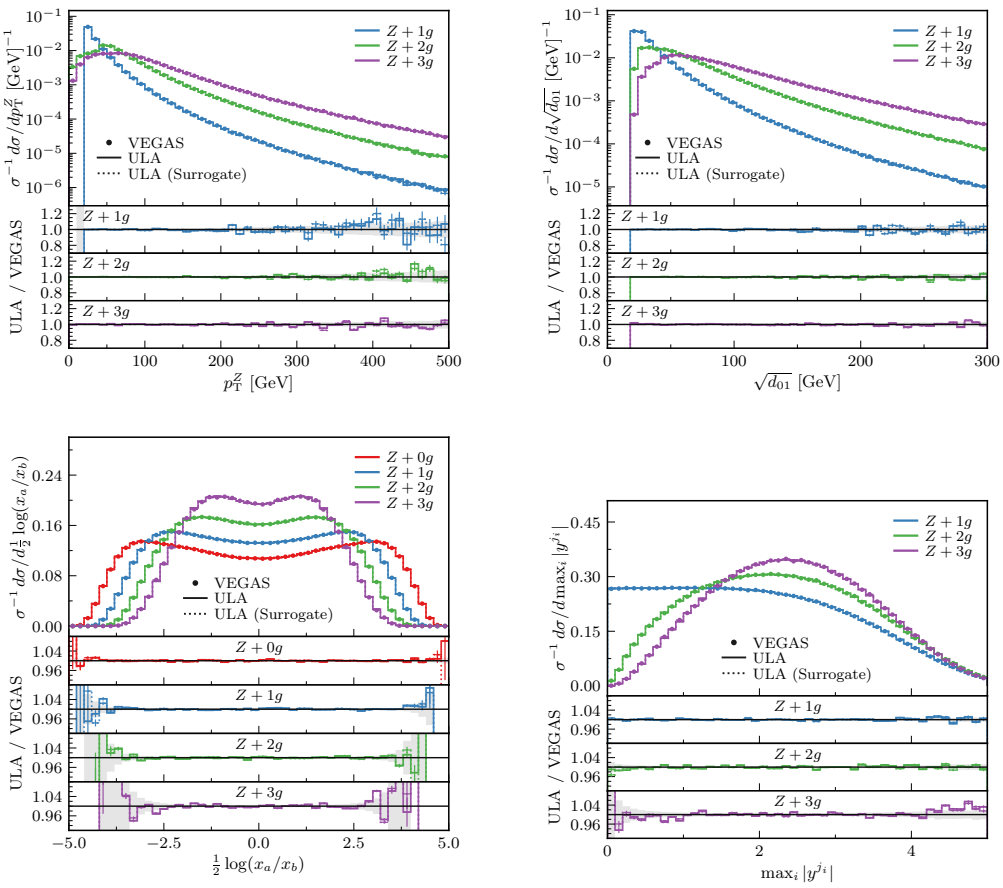


Figure 2: Normalized differential distributions for the Z-boson transverse momentum, the exclusive k_t clustering scale $\sqrt{d_{01}}$, the full-system boost rapidity $\frac{1}{2}\log(x_a/x_b)$, and the largest absolute gluon rapidity $\max_i |y^{j_i}|$. Filled markers show the VEGAS reference, solid lines show ULA samples, and dotted lines show ULA samples after surrogate initialization. The lower panels show the corresponding ratios to VEGAS for each multiplicity.

4 Discussion

We illustrated that parallel Langevin sampling with a learned Stein diagnostic offers a practical route to direct unit-weight event generation. For the $Z+$ jets processes studied here, relaxation times grow more mildly with gluon multiplicity than VEGAS inverse unweighting efficiencies and are further reduced by neural surrogate initialization.

Our method requires gradient evaluations, whose cost affects the runtime. Reverse-mode automatic differentiation requires one forward and backward pass, so for tree-level matrix elements composed of rational functions, gradient evaluation should approach a small constant-factor overhead over regular evaluation [78, 79].

The present study considers a fixed partonic channel, tree-level matrix elements, and a simple phase-space map. Establishing the scaling at higher multiplicity and comparing with production methods requires implementation in a full event generator. The parallel chains, convergence diagnostic, and surrogate initialization provide a concrete starting point for that study.

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A Underdamped Langevin algorithm

In this appendix, we describe the Underdamped Langevin algorithm implementation in detail. Each MCMC chain carries a position x and a velocity v. We assume we have access to the score $s(x) = \nabla \log \widetilde{p}(x)$. For stability purposes the score is regularized component-wise to $\widetilde{s}_i(x) = s_i(x)/(1 + \eta_i|s_i(x)|)$, where η_i are the MCMC step sizes [80]. At time step t, a new proposal x', v' is generated following the leapfrog step

$$\begin{aligned} v_{1/2} &= v_t + \frac{1}{2}\Lambda_\eta\, \widetilde{s}(x_t), \\ x' &= x_t + \Lambda_\eta\, v_{1/2}, \\ v' &= v_{1/2} + \frac{1}{2}\Lambda_\eta\, \widetilde{s}(x'), \end{aligned} \tag{8}$$

where $\Lambda_\eta = \mathrm{diag}(\eta_1, \ldots, \eta_d)$. This step discretizes the Hamiltonian dynamics on the augmented density

$$\pi(x, v) \propto \widetilde{p}(x) \exp\left(-\frac{1}{2} v^T v\right). \tag{9}$$

In the continuous-time limit, the Hamiltonian flow conserves the Hamiltonian $H(x, v) = -\log \widetilde{p}(x) + v^T v/2$ and preserves phase-space volume. As a result, $\pi(x, v)$ as well as its x-marginal, which is proportional to the desired target density $\widetilde{p}(x)$, is invariant. However, at finite step sizes an integration error is introduced. By applying a MH acceptance step with probability

$$a = \min\left\{1, \exp\left[-H(x', v') + H(x_t, v_t)\right]\right\} \tag{10}$$

the discretization bias is corrected. Upon rejection, the position is kept and the incoming momentum is reversed $v_t \to -v_t$. After the MH step, the momentum is refreshed

$$v_{t+1} = \beta v_{\mathrm{acc}} + \sqrt{1 - \beta^2}\, \zeta, \qquad \zeta \sim \mathcal{N}(0, I). \tag{11}$$

These momentum updates preserve the Gaussian velocity marginal and inject randomness into the otherwise deterministic Hamiltonian evolution, allowing the chain to move between trajectories and explore the full target distribution. After equilibration, terminal positions drawn from these chains are distributed according to $\widetilde{p}(x)$.

We write the step sizes as $\log \eta_i = \log \epsilon + \log \rho_i$, with $\sum_i \log \rho_i = 0$. The scalar scale ϵ is initialized by doubling or halving until the average acceptance probability is close to one half [38], while the directions ρ_i are initialized from the diagonal score second moment, motivated by Fisher preconditioning [81, 82],

$$F_{ii,t} = \mathbb{E}_{x_t}\left[s_i(x_t)^2\right], \qquad \log \rho_i = -\frac{1}{2}\log F_{ii,t} + \frac{1}{2d}\sum_{j=1}^{d} \log F_{jj,t}. \tag{12}$$

This fixes $\prod_i \rho_i = 1$ and increases step sizes in directions with smaller typical score variation. During the chain, $\log \epsilon$ is adapted by ascending the expected squared jump distance $\mathrm{ESJD}(\epsilon) = \mathbb{E}_{x_t, v_t, \zeta}\left[a\|x' - x_t\|^2\right]$ at fixed ρ, a standard proxy for MCMC efficiency [83, 84]. The directions are adapted by updating ρ_i with new estimates of the diagonal score second moment. Both the ESJD scale update and the score-moment direction update use the same diminishing Robbins–Monro schedule, so that the transition-kernel changes vanish asymptotically, as required for stationarity [85, 86]. All expectations are evaluated over the parallel chain ensemble.

B Stein diagnostic validation

In this appendix, we validate the Stein diagnostic on a target for which the MCMC convergence can be computed analytically. Starting from a fixed point μ_0 away from some target Gaussian mean μ_q and diagonal target Σ_q, we sample an exact Ornstein–Uhlenbeck (OU) transition kernel

$$x_{t+1} = \mu_q + e^{-\eta\Sigma_q^{-1}}(x_t - \mu_q) + \Sigma_q^{1/2}\left(I - e^{-2\eta\Sigma_q^{-1}}\right)^{1/2}\xi, \qquad \xi \sim \mathcal{N}(0, I), \tag{13}$$

where η is the step size. After t steps, the marginal distribution is Gaussian with

$$\mu_t = \mu_q + e^{-t\eta\Sigma_q^{-1}}(\mu_0 - \mu_q)$$
$$\Sigma_t = \Sigma_q \left(I - e^{-2t\eta\Sigma_q^{-1}}\right). \tag{14}$$

Thus p_t converges to the target density as $t \to \infty$, yielding a controlled and testable sequence of distributions. To validate the role of the boundary function, we apply a radial cut that removes 20% of the target probability mass centered around the target mean μ_q.

Under Eq. (5), the optimal Stein discrepancy is [48]

$$S_{\text{opt}}(p_t, q) = \frac{1}{2\lambda}\mathbb{E}_{x\sim p_t}\left[\|s_q(x) - s_{p_t}(x)\|^2\right]. \tag{15}$$

We compare this optimal value against the LSD computed on OU samples and on ULA samples. The validation results use dimensions $d = 4$ and $d = 16$, with target parameters

$$\mu_q = \text{linspace}(-2, 2, d), \qquad \Sigma_q = \text{diag}\big(\text{linspace}(0.5, 2, d)^2\big). \tag{16}$$

The OU process is initialized from the fixed point $\mu_0 = \mu_q + 3$, while ULA is initialized from a unit Gaussian. The LSD regularization is $\lambda = 0.1$. The Stein critic is modelled with a five-layer residual GeLU network with hidden width 128. Optimization is performed with the Adam optimizer [87] on $n = 10^7$ chains, and final LSD values are obtained using 10-fold cross validation. The boundary function is $h(x) = \max\{1 - r_{\text{cut}}^2/r^2(x), 0\}$.

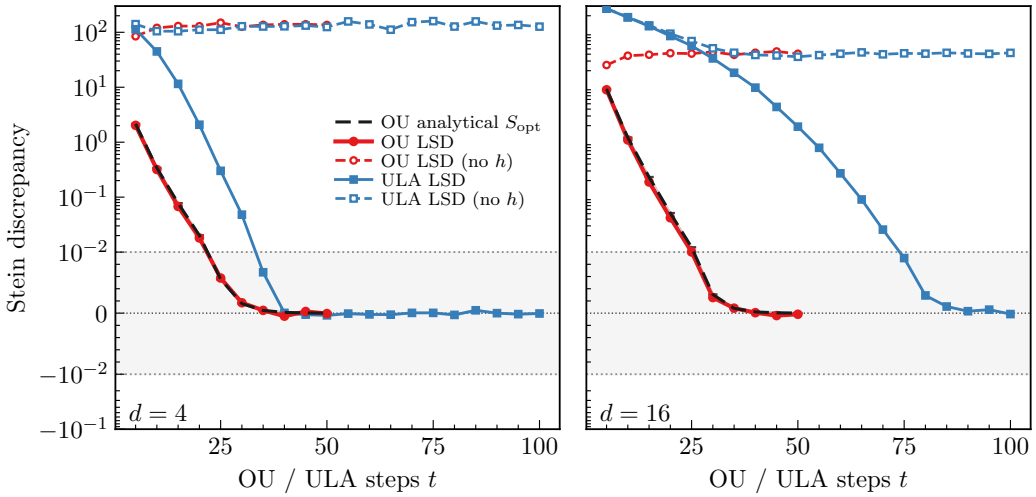


Figure 3: The LSD as a function of the number of steps for the exact OU reference process with $\eta = 0.5$ (red) and the underdamped Langevin algorithm (ULA, blue). Dashed lines show the boundary-function ablation. The black line shows the analytic optimal Stein discrepancy S_{opt}.

The validation results are shown in Fig. 3. For the exact OU samples, the learned LSD tracks the analytic value S_{opt}, supporting its use as a quantitative convergence diagnostic. The ULA curves show similar monotonic decay towards zero, with the higher-dimensional case converging more slowly, qualitatively consistent with the expected $\sqrt{d}$ scaling of Metropolis-adjusted Langevin methods [88]. In the absence of the boundary function, the remaining finite boundary contribution prevents the Stein discrepancy from approaching zero.